

MOHAMMED AZEEZULLA

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience building and deploying machine learning systems, Generative AI pipelines, and agentic workflows across AWS, GCP, and Azure. Hands-on background in LLM orchestration, RAG architecture, multi-agent systems, and MLOps - delivering production-ready solutions with measurable gains in latency, retrieval quality, and system reliability.

TECHNICAL SKILLS

- **Generative AI:** LangChain, LangGraph, LlamaIndex, AutoGen, CrewAI, MCP, RAG, Embeddings, Prompt Engineering-, Fine-tuning (LoRA, PEFT), LLM Evaluation, Guardrails
- **ML & NLP:** HuggingFace, Transformers, Supervised/Unsupervised Learning, NLP, Model Optimization, Feature Engineering, Time Series, Recommendation Systems
- **MLOps:** Docker, Kubernetes, Terraform, MLflow, CI/CD, Model Versioning, Feature Stores, Monitoring & Drift Detection
- **Cloud:** AWS (SageMaker, Bedrock, Lambda, ECS, S3, BentoML), GCP (Vertex AI, BigQuery, Cloud Run), Azure (Azure ML, Functions)
- **Data & Databases:** Apache Spark, Kafka, ETL Pipelines, FAISS, Pinecone, Weaviate, SQL, NoSQL
- **Languages & Tools:** Python (Advanced), SQL, JavaScript, FastAPI, Flask, REST APIs, Git, Linux

PROFESSIONAL EXPERIENCE

JPMorgan Chase & Co | *AI/ML Engineer*

Apr 2025 - Present

- Developed and deployed Generative AI applications using LlamaIndex, improving retrieval precision by ~28% measured via automated evaluation benchmarks
- Built RAG-based pipelines with FAISS and Pinecone to enhance contextual retrieval accuracy and reduce hallucination rates in production query responses
- Architected multi-agent AI systems using AutoGen to automate document processing and data extraction workflows, reducing manual effort across operations teams
- Deployed inference services using AWS SageMaker and BentoML, reducing average response latency from ~1.8s to ~1.3s through endpoint optimization and request batching
- Designed end-to-end ML deployment pipelines using Kubernetes, Docker, and Terraform, enabling reproducible and scalable model releases
- Established MCP pipelines to unify AI agents with enterprise applications and internal data repositories.

HCLTech | *AI Engineer*

Jun 2022 - Nov 2023

- Trained and fine-tuned HuggingFace transformer-based NLP models for text classification and named entity recognition, improving F1 score by ~18% over the BERT baseline
- Built embedding-based semantic search pipelines using Sentence Transformers and FAISS, integrated into internal knowledge retrieval tools ahead of broader LLM adoption
- Deployed NLP models on Azure ML with managed endpoints, enabling scalable real-time inference and automated performance monitoring
- Developed agent-based automation workflows using CrewAI for document processing and support operations.
- Implemented LangGraph based pipelines for stateful multi-step data extraction and workflow optimization.

Tech Mahindra | *ML Engineer*

Dec 2020 - May 2022

- Engineered supervised ML models for customer churn prediction, product classification, and collaborative filtering-based recommendation
- Trained and deployed models on GCP AI Platform (now Vertex AI) using managed training jobs and hyperparameter tuning pipelines
- Built data preprocessing and feature engineering workflows in Python and Apache Spark for large-scale structured datasets
- Created FastAPI-based microservices to expose ML model predictions as REST endpoints consumed by internal business applications

KEY PROJECTS

Generative AI Multi-Agent System

- Built an autonomous task execution system using LangGraph and CrewAI with shared memory and tool-use across agents for multi-step reasoning
- Integrated RAG pipelines for context injection, enabling agents to retrieve and synthesize domain-specific information at inference time

LLM-Powered Knowledge Assistant

- Developed a production-grade RAG chatbot using LangChain, OpenAI embeddings, and Pinecone, with response reranking to improve answer relevance
- Implemented chunking strategies and hybrid retrieval (dense + sparse) to reduce hallucination and improve factual grounding

EDUCATION

DePaul University | *M.S., Artificial Intelligence*

Nov 2025